

SMART FARMER
AI INTEGRATED SPARE PART LIFECYCLE FORECASTING SYSTEM

Spare Part Image Recognition and 3D Visualization System

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DECLARATION OF THE CANDIDATE

I declare that this is my own work and this report does not incorporate, without acknowledgement, any material previously submitted for a degree or diploma in any other university or institution of higher learning. To the best of my knowledge and belief, it does not contain any material previously published or written by another person except where acknowledgement is made in the text.

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ABSTRACT

Image recognition and 3D visualization play an increasingly important role in improving efficiency and accuracy in industrial and mechanical systems. In complex machinery environments, identifying specific spare parts and understanding their exact location can be challenging, especially for users without experiences. This research addresses this issue by proposing an intelligent system that integrates an image recognition with interactive 3D visualization to support spare part identification and analysis. The system is designed to recognize machine components from images and visually represent them within a 3D model, providing a clearer understanding of their structure and placement.

A dataset of machine spare part images was collected under different conditions and angles to ensure robustness against variations such as lighting, angle, and background noise. A deep learning model based on ResNet-50 was utilized and fine-tuned using transfer learning to achieve high classification accuracy while maintaining computational efficiency. The trained model processes input images and predicts the corresponding spare part along with a confidence score.

To enhance usability, the identified component is mapped to a 3D model of the machine using modern web-based visualization technologies. The relevant part is dynamically highlighted, allowing users to interact with the model through rotation, zooming, and detailed exploration. In addition, AI-based video generation is incorporated to visually demonstrate the position and functionality of the detected component, further improving user understanding.

The proposed system provides an effective solution for applications such as maintenance support, training, and spare part identification. Experimental outcomes demonstrate that the integration of image recognition and 3D visualization significantly improves both accuracy and user experience. This approach establishes a scalable foundation that can be extended to more complex machinery and advanced predictive maintenance systems in the future.

Keywords Image recognition, 3D visualization, ResNet-50, deep learning, transfer learning, Spare part Identification

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LIST OF ABBREVIATIONS

AI	– Artificial Intelligence
CNN	– Convolutional Neural Network
ResNet	– Residual Neural Network
DNN	– Deep Neural Network
IoT	– Internet of Things
GPU	– Graphics Processing Unit
TfLite	- Tensor Flow Lite
ML	– Machine Learning
API	– Application Programming Interface
RGB	– Red Green Blue
JSON	– JavaScript Object Notation
UI	– User Interface
UX	– User Experience
CAD	– Computer Aided Design
AR	– Augmented Reality

CHAPTER 1: INTRODUCTION

Modern industrial and mechanical systems have become increasingly complex due to rapid technological advancements and the integration of sophisticated components. As a result, the identification, maintenance, and management of machine spare parts have become more challenging tasks, particularly for individuals without specialized technical expertise. In industries such as agriculture, manufacturing, and heavy machinery operations, accurate identification of components is crucial to ensure proper maintenance, reduce downtime, and avoid costly breakdowns. However, the traditional approach to spare part identification relies heavily on expert knowledge, manuals, and physical inspection, which can be time-consuming, error-prone, and inefficient.

In many real-world scenarios, users such as technicians, operators, or trainees may struggle to recognize specific components due to similarities in shape, size, and structure among different parts. Incorrect identification can lead to improper maintenance decisions, delays in repair processes, and increased operational costs. Furthermore, understanding the exact location of a component within a complex machine structure is equally important, as it directly impacts the effectiveness of maintenance and replacement procedures. Therefore, there is a growing need for intelligent systems that can assist users in both identifying and locating machine spare parts accurately and efficiently.

With the advancement of computer vision, image processing, and deep learning technologies, automated object recognition systems have gained significant attention in recent years. These systems are capable of analyzing visual data and extracting meaningful information, enabling accurate classification and detection of objects across various domains. Deep learning models, particularly Convolutional Neural Networks (CNNs), have demonstrated remarkable performance in image classification tasks due to their ability to learn complex patterns and features from large datasets. Among these architectures, ResNet-50 has proven to be highly effective, as it addresses the vanishing gradient problem and enables the training of deeper networks with improved accuracy.

Despite the success of existing image recognition systems, most of them focus solely on classification and provide limited support for practical applications. In the context of spare part identification, simply predicting the name of a component is not sufficient. Users also require a clear understanding of where the part is located within the machine and how it is structured. Without this spatial awareness, the usefulness of such systems remains limited, particularly for training and real-world maintenance tasks. Additionally, many existing solutions lack interactivity and fail to provide an intuitive user experience, which reduces their effectiveness for non-expert users.

To overcome these limitations, this research proposes an integrated approach that combines image recognition with interactive 3D visualization for enhanced spare part identification. The

primary objective of this study is to develop a system that not only identifies machine components from images but also visually represents their location within a 3D model of the machine. This integration aims to bridge the gap between intelligent detection and user-friendly visualization, providing a more comprehensive solution for real world applications.

In this research, a dataset of machine spare part images was collected under varying environmental conditions, including different lighting settings, angles, and backgrounds, to ensure robustness and generalization of the model. A deep learning model based on ResNet-50 was utilized and fine-tuned using transfer learning techniques. Transfer learning enables the use of pre trained weights from large scale datasets, significantly reducing training time while maintaining high accuracy. The model was trained to classify different spare parts and generate predictions along with confidence scores, ensuring reliability in the identification process. Following the recognition phase, the identified component is mapped to a corresponding part within a 3D model of the machine. The visualization component is developed using modern web-based 3D technologies, enabling real time rendering and interaction. The detected part is dynamically highlighted within the model, allowing users to explore the structure through rotation, zooming, and navigation. This interactive approach provides a clear and intuitive understanding of the spatial arrangement of machine components, which is essential for maintenance and training purposes.

Furthermore, to enhance user engagement and comprehension, AI-based video generation techniques are incorporated into the system. These videos provide animated demonstrations of the identified spare part within the machine, illustrating its position, orientation, and potential function. This additional layer of visualization improves knowledge transfer and makes the system more accessible to users with limited technical background.

The proposed system offers several advantages over traditional methods and existing solutions. It reduces dependency on expert knowledge, minimizes errors in spare part identification, and improves efficiency in maintenance processes. Additionally, the combination of image recognition and 3D visualization enhances the overall user experience by providing both accurate detection and meaningful visual feedback. The system is also designed to be scalable, allowing future expansion to include more complex machinery, additional components, and advanced features such as predictive maintenance.

In conclusion, this research aims to develop an intelligent and interactive solution for spare part identification by leveraging the power of deep learning and 3D visualization technologies. By integrating accurate image recognition with immersive visual representation, the proposed system addresses key challenges in modern mechanical systems and provides a practical tool for maintenance support, technical training, and component management. This work establishes a strong foundation for future research and development in the field of intelligent diagnostic systems and smart industrial applications.

1.1. Background and Literature Survey

The importance of image recognition and 3D visualization for spare part identification has been highlighted in the introduction section. This section reviews existing research related to computer vision, deep learning, and visualization technologies relevant to this study. The structure follows similar approaches used in previous literature surveys.

Computer vision has been widely used for automated object identification in various domains. Traditional image processing methods relied on manually extracted features such as color, shape, and texture, but these methods often produced low accuracy under varying environmental conditions [1]. To overcome these limitations, deep learning techniques, particularly Convolutional Neural Networks (CNNs), have been introduced and have shown significant improvements in image classification tasks [2].

CNN-based models are capable of automatically learning complex visual features, making them more effective than traditional machine learning methods such as Support Vector Machines [3]. With the advancement of deep learning, transfer learning has become a popular approach, allowing pre trained models to be fine-tuned for specific tasks with limited datasets [4].

Among various CNN architectures, ResNet-50 has demonstrated high performance in image recognition tasks due to its deep structure and residual connections, which help overcome training issues such as vanishing gradients [5]. Several studies have shown that ResNet-50 performs well under real-world conditions, making it suitable for identifying machine spare parts captured from different angles and environments [6].

Furthermore, AI-based visualization techniques, including video generation, have been introduced to enhance user understanding by providing animated demonstrations of components and their functions [7]. However, most existing systems focus only on classification or visualization separately, without integrating both aspects into a single solution [8].

Therefore, this research proposes an integrated approach that combines ResNet-50-based image recognition with interactive 3D visualization and AI-generated video support. This approach aims to improve both accuracy and user understanding by not only identifying spare parts but also visually representing their location within a 3D model, providing a more practical and user-friendly solution.

1.2. Problem Statement

The central problem that this research addresses can be stated as follows: there is no automated, accessible, and accurate tool available to smallholder farmers in Sri Lanka or in comparable developing agricultural economies to visually assess the wear condition of their machinery's spare parts in a way that is both granular enough to be actionable and simple enough to be used in a field setting.

Manual inspection, the current standard, is unreliable for several interconnected reasons. First, the threshold for 'acceptable wear' varies from one inspector to another. A farmer who is used to running equipment hard may consistently underestimate wear severity, while an overly cautious mechanic may recommend premature replacement, adding unnecessary cost. Second, the knowledge required to accurately read the signs of part degradation is specialist knowledge that many smallholder farmers simply do not possess. A pattern of subtle surface cracking on a V-belt, for example, may be overlooked by an untrained eye but represents a part that is weeks away from failure [9].

Third, and perhaps most critically, manual inspection is a whole-part assessment. A mechanic will look at a tyre and say 'this looks worn' without necessarily identifying which part of the tyre is worn, by how much, and in what pattern. This lack of spatial precision in the assessment means that the information produced cannot be used for meaningful predictive modelling. To forecast when a part will fail, the system needs to know not just that damage exists, but where the damage is, how large it is, and how it relates to the critical load-bearing or functional surfaces of the component [10].

Existing mobile applications for spare part identification, such as those built on whole image classification models, have shown that AI-assisted visual inspection is conceptually viable in the agricultural domain [11]. However, applications built on standard image classifiers which assign a single label to an entire image do not address the spatial dimension of the problem. They can tell you that an image shows a 'damaged tyre', but they cannot show you exactly where the bald patch is, how large it is relative to the tyre's total surface, or whether it overlaps with the high stress contact patch. This spatial information is precisely what is needed to compute meaningful wear metrics and integrate them into a lifecycle forecasting model [12].

A further complication in this specific research context is that publicly available datasets of agricultural spare part wear are extremely limited. Most computer vision benchmarks are built on consumer or industrial contexts automotive assembly lines and do not reflect the visual characteristics of agricultural equipment wear as it appears in the field conditions of rural Sri Lanka. The mud, uneven lighting, varied backgrounds, and scale differences that characterize on-farm photography are significantly different from the clean, controlled images used to train most published models [13]. This dataset gap represents a foundational challenge and is the reason why the data collection methodology described in this report places such heavy emphasis on sourcing images from real local agricultural settings.

1.3. Research Objectives

The central problem addressed in this research can be list as follows: there is a lack of an integrated, intelligent, and user-friendly system that can both accurately identify machine spare parts from images and provide a clear understanding of their spatial location within a machine through interactive visualization. While image recognition technologies have advanced significantly, most existing solutions remain limited in practical usability, especially for new users with less experience in real world environments.

Current practices for spare part identification are largely manual and depend heavily on user experience and technical knowledge. In many cases, technicians or operators rely on visual inspection and reference manuals to identify components. However, this process is often unreliable due to similarities in the appearance of parts, variations in lighting conditions, and differences in user expertise. As a result, incorrect identification may occur, leading to delays in maintenance and increased operational costs [1].

Although several computer vision-based systems have been developed for object classification, most of these systems rely on whole-image classification approaches. These models are capable of assigning a label to an image but lack the ability to provide contextual or spatial information about the identified component [2]. For example, a model may correctly classify an image as a specific spare part, but it does not assist the user in understanding where that part is located within the overall machine structure or how it is connected to other components. This limitation reduces the effectiveness of such systems in real-world maintenance and training applications.

Furthermore, many existing applications focus solely on recognition accuracy without considering user interaction and visualization. The absence of interactive features makes it difficult for users, especially beginners, to fully understand the structure and placement of machine components. Without visual guidance, users may still require expert assistance even after obtaining classification results [3].

Another major limitation in this research area is the lack of domain specific datasets for machine spare parts. Most publicly available datasets are designed for general object recognition tasks and do not include sufficient variations of mechanical components under real-world conditions. Images captured in practical environments often with less lighting, complex backgrounds, varying scales, and different angles, which are not well represented in standard datasets [4]. This gap highlights the need for collecting and training models on realistic data to ensure robustness and accuracy.

Additionally, while 3D visualization technologies have been widely used in engineering and simulation, they are rarely integrated with image recognition systems. Existing 3D systems mainly function as standalone visualization tools without intelligent detection capabilities. Similarly, image recognition systems operate independently without providing immersive visual feedback [5]. This separation creates a gap in developing a unified system that combines both intelligent recognition and interactive visualization.

Moreover, the use of AI-based video generation for enhancing technical understanding is still in its early stages within this domain. Although such techniques have the potential to improve user learning and engagement, they are not commonly applied in spare part identification systems [6]. Therefore, this research aims to address these limitations by proposing an integrated solution that combines ResNet-50 based image recognition with interactive 3D visualization and AI-generated video support. By bridging the gap between accurate detection and intuitive visualization, the proposed system provides a more practical, scalable, and user-friendly approach for spare part identification, maintenance support, and technical training.

1.4. Scope of the Research

The scope of this research is defined by the following parameters:

- **Component Types:** The research focuses on identifying selected machine spare parts used in the system dataset. Other complex or internal components are acknowledged but remain outside the scope of this study.
- **Functionality:** The system is limited to image-based classification of spare parts using a deep learning model based on ResNet-50. It does not perform real time object detection or defect analysis.
- **Visualization:** The study includes interactive 3D visualization, where identified spare parts are highlighted within a pre-developed 3D machine model. Full simulation or physics-based modeling is not considered. It helps users to get a knowledge on how the spare part works, where does it locates and etc.
- **Platform:** The system is designed as a web-based or mobile-compatible application, ensuring accessibility and ease of use. Performance is optimized for standard devices rather than high end hardware.

- **Dataset:** The model is trained using a custom dataset of spare part images collected under different conditions such as varying lighting, angles, and backgrounds. The dataset size and variety are limited to the scope of this research.
- **Geographical Context:** Data collection and system applicability are primarily based on local conditions, making the system more suitable for similar environments, although the methodology can be extended globally.
- **Training Environment:** Model training is conducted using GPU-supported environments, while deployment is designed for lightweight inference on user devices or web platforms.

1.5. Justification and Significance

The significance of this research arises from both its practical applicability and its contribution to the evolving field of computer vision integrated with 3D visualization technologies. From a practical perspective, the proposed system provides an efficient solution for identifying machine spare parts and understanding their spatial location within complex machinery. In real-world scenarios such as maintenance and repair, incorrect identification of components can lead to operational delays, increased costs, and system failures. By enabling accurate image-based recognition and interactive 3D visualization, this research helps reduce dependency on expert knowledge and improves the efficiency of maintenance processes [7].

From a technical standpoint, this research contributes to the integration of deep learning and visualization technologies into a unified system. While previous studies have focused mainly on image classification or object detection, this work combines a model based on ResNet-50 with interactive 3D visualization to provide both identification and spatial understanding for help users to identify and gain a knowledge about the task of the spare part. The use of transfer learning improves model performance under real-world conditions such as varying lighting, angles, and backgrounds. Additionally, the development of a custom dataset of machine spare parts adds value to this research, as such domain-specific datasets are limited and not widely available [8].

Another key contribution of this research is the inclusion of AI-based video generation to enhance user understanding. Instead of relying solely on static outputs such as labels or images, the system provides dynamic visual explanations highlighting the position and structure of the identified component. This approach improves knowledge transfer and makes the system more accessible to non-technical users.

The broader significance of this research lies in improving accessibility to advanced technological solutions in mechanical and industrial domains. By developing a system that can be deployed on mobile platforms, advanced tools such as automated identification and interactive visualization are made available to a wider range of users, including farmers, mechanics, and shop keepers who sell spare parts [9]. This contributes to the democratization of intelligent maintenance systems, where users can benefit from AI-driven assistance without requiring extensive technical expertise.

Furthermore, this research establishes a scalable foundation for future developments such as predictive maintenance, real-time detection, and digital twin integration. By demonstrating that accurate spare part recognition combined with immersive visualization is achievable using practical datasets and efficient models, this study supports the advancement of smart maintenance systems and intelligent industrial applications.

CHAPTER 2: METHODOLOGY

2.1. Overview of the Research Methodology

The methodology followed in this research was structured as a sequential and systematic process, integrating image recognition with 3D visualization to achieve accurate spare part identification and display the correct location of its location. The overall approach was designed to ensure reliability, scalability, and practical usability in real-world conditions. A deep learning-based classification model and interactive visualization techniques were combined to form a unified system, with each stage carefully planned and executed.

The methodology was primarily focused on developing and evaluating an image classification model based on ResNet-50, followed by mapping the identified components to a 3D model and enhancing user understanding through visualization and AI-generated video content. The workflow was designed to be reproducible and adaptable for future extensions.

The following major stages constituted the complete research workflow:

- **Identification of Target Spare Parts:**
Relevant machine components were selected based on system requirements and availability of data.
- **Dataset Collection:**
Real-world images of spare parts were collected under different conditions such as varying lighting, angles, and backgrounds to ensure robustness.
- **Data Preprocessing and Preparation:**
Images were cleaned, resized, labeled, and augmented to improve model generalization and performance.
- **Model Selection and Training:**
A ResNet-50 based deep learning model was selected and fine-tuned using transfer learning techniques for efficient and accurate classification.
- **Model Evaluation:**
The trained model was evaluated using performance metrics such as accuracy and validation loss to ensure reliability.

- **3D Model Development:**
A digital 3D model of the machine was created using modern visualization technologies to represent components structurally.
- **Integration of Recognition and Visualization:**
The predicted spare part was mapped to the corresponding component in the 3D model, where it was dynamically highlighted.
- **System Integration and Deployment Design:**
The overall system was designed for deployment as a mobile application, ensuring accessibility and usability.

Each of these stages was implemented carefully to ensure that the system not only achieves accurate spare part recognition but also provides meaningful visual feedback, thereby improving user understanding and practical application.

2.2. Data Collection from Real World Agricultural Sources

2.2.1. Rationale for primary data collection

The decision to collect primary data was based on the limitations of existing public datasets. Most available computer vision datasets are designed for general object recognition and are collected under controlled conditions, which do not reflect real-world environments where spare part images are captured [10]. These datasets often lack variations in lighting, background, and viewing angles, making them less suitable for practical applications.

In addition, publicly available datasets rarely include locally available machine spare parts. Variations in design, size, and condition of components found in local markets are not represented in standard datasets, which can affect model accuracy [11]. To address this gap, images and catalogues were collected directly from local spare part shops under professionals.

This primary data collection approach ensured that the dataset reflects real-world conditions and local component variations. As a result, the trained model becomes more robust, reliable, and suitable for practical deployment in spare part identification systems.

2.2.2. Data collection from local sources

Data collection for this research was conducted through visits to local spare part shops and related sources, primarily located in the Western Province of Sri Lanka. Prior to collecting data, permission was obtained from shop owners, and the purpose of the research was clearly explained. The data collection process was carried out over several weeks to ensure sufficient coverage of different spare parts and conditions.

Images of machine spare parts were captured in real-world environments, including shop displays, storage areas, and catalogues. Although controlled environments could produce clearer images, they would not represent the actual conditions under which users capture images. Therefore, capturing images in natural lighting and varied backgrounds ensures that the trained model performs reliably during real-world usage [12].

Images were taken using different smartphone cameras to simulate typical user devices. Each spare part was captured from multiple angles, distances, and lighting conditions to increase dataset diversity. In addition to photographs, catalogues obtained from local shops were also used to supplement the dataset, providing clear reference images of spare parts.

Furthermore, parts were collected in brand new format. This diversity helps improve the robustness and generalization ability of the deep learning model. Overall, a substantial number of images were gathered to support effective training and evaluation of the system.

Table 1 Summary of some image dataset by catalogue ID and Part name:

Catalogue ID	Part Name	No. of Images
TA202-2060-0	Pressure Plate	40
T1150-2017-6	Clutch Plate	48
3C001-2635-0	Bearing Release	37
TA040-26723	Main Wheel Axel	51
TC222-37432	Lift Arm Shafter	48
HH164-32430	Oil Filter	32
HHTA0-37710	HST Filter	46
32440-26830	Axel Wheel	45
HH15221-43170	Fuel Filter	52
TA040-23702	Shaft PTO	48
TC230-14630	4Wheel Shafter	53
7C505-55232	Holder Bearing	43

TD030-1201-0	Assy Gear Bevel	60
7P038-5120-0	Assy Gear Bevel	60
35080-44662	Joint Tie Rod	42
TD030-13710	End Rod	56
7P038-53112	Plate	44
IA021-73030	Assy Pump Water	52
1A091-04362	Gasket Bearing Case	20
Total		902

2.3. Data Preprocessing and Annotation

2.3.1. Image quality screening

Following data collection, all raw images were subjected to a manual quality screening process to ensure dataset reliability. Images were excluded based on several criteria, including severe motion blur, poor focus, extreme overexposure or underexposure, and cases where the spare part was not clearly visible or occupied only a small portion of the frame. Additionally, near duplicate images with no significant variation in angle or lighting were removed. This filtering process ensured that only high quality and meaningful images were retained for model training and evaluation.

2.3.2. Annotation methodology

Since this research focuses on image identification rather than object detection, the annotation process was performed at the image level. Each image was assigned a single class label representing the corresponding spare part type. Annotation was carried out using a structured labeling approach to ensure consistency across the dataset.

A clear set of labeling guidelines was defined prior to annotation to avoid ambiguity. These guidelines ensured that each spare part category was consistently labeled based on its visual characteristics such as shape, structure, and design. Unlike object detection tasks, no bounding boxes or region-based annotations were required, as the entire image was considered for classification.

To maintain annotation quality, a verification step was performed where labeled images were reviewed to ensure correctness and consistency. Any ambiguous or incorrectly labeled samples were corrected to prevent errors during model training.

2.3.3. Dataset splitting

The prepared dataset was divided into three subsets: training, validation, and testing. A split ratio of 50% for training, 25% for validation, and 25% for testing was adopted. This distribution was selected to ensure sufficient data for model learning while maintaining reliable evaluation. The dataset split was performed in a balanced manner so that each spare part category was proportionally represented across all subsets.

2.3.4. Data augmentation

To improve model generalization and performance, data augmentation techniques were applied to the training dataset. These techniques artificially increased dataset diversity by introducing variations similar to real-world conditions. The following augmentation operations were applied:

- Horizontal flipping to simulate different viewing directions.
- Random rotation within a defined range to represent varied image capture angles.
- Brightness and contrast adjustments to mimic different lighting conditions.
- Random zoom to simulate varying distances from the object.
- Minor positional shifts to reflect off-center image capturing.

These augmentations helped the model become more robust to variations in input images and reduced the risk of overfitting. Since this research uses a classification approach, augmentation was applied directly to images without requiring coordinate transformations.

2.4. ResNet-50 Classification Approach

2.4.1. Rationale for ResNet-50

The initial phase of this research selected a deep learning architecture based on ResNet-50 for image classification. ResNet-50 is a widely used convolutional neural network architecture known for its ability to train very deep networks effectively by introducing residual connections. These residual connections allow the network to learn identity mappings, helping to overcome the vanishing gradient problem that typically affects deep neural networks [13].

Unlike traditional convolutional networks, ResNet-50 uses skip connections that enable information to flow directly across layers. This design improves feature learning and allows the model to capture complex patterns such as shapes, textures, and structural variations of machine

spare parts. The architecture consists of 50 layers, providing a strong balance between computational efficiency and high classification accuracy. For this research, ResNet-50 was selected due to its proven performance in real-world image classification tasks and its ability to generalize well across varying conditions.

2.4.2. Transfer learning with ResNet-50

Rather than training the model from scratch, transfer learning was employed to improve efficiency and performance. The ResNet-50 model was initialized with weights pre-trained on the ImageNet dataset, which contains millions of images across a wide range of object categories [14]. This allows the model to leverage previously learned low-level and high-level visual features such as edges, textures, and shapes.

The transfer learning process followed these steps:

- Load the ResNet-50 base model with pre-trained ImageNet weights, excluding the original classification layer.
- Freeze the convolutional base layers to preserve learned feature representations.
- Add a new classification head consisting of:
 - Global Average Pooling layer
 - Dropout layer (rate = 0.3) to reduce overfitting
 - Dense layer with SoftMax activation for classification
- Train the classification head for initial epochs to adapt to spare part categories
- Unfreeze the top layers of the network and perform fine-tuning with a lower learning rate to improve performance on domain-specific features

This approach allowed efficient training while maintaining high accuracy with a relatively smaller dataset.

2.4.3. ResNet-50 training configuration

The following configuration was used during model training:

- Input image size: $224 \times 224 \times 3$ (RGB)
- Batch size: 32
- Optimizer: Adam
- Initial learning rate: $1e-4$
- Fine-tuning learning rate: $1e-5$
- Loss function: Categorical cross-entropy

- Evaluation metric: Accuracy
- Early stopping: Based on validation loss (patience = 5 epochs)
- Model checkpoint: Best model saved based on validation performance

Training was performed using a GPU-enabled environment, and the model was implemented using TensorFlow with the Keras framework.

2.4.4. ResNet-50 inference pipeline

During inference, the input image is first resized to 224×224 pixels and normalized to a range of $[0, 1]$. The processed image is then passed through the trained ResNet-50 model. ResNet-50 outputs an integer ID for each part. By that integer part system checks the relevant details from the database similar to that ID.

The output layer generates a probability distribution across all defined spare part classes using the SoftMax function. The class with the highest probability is selected as the predicted spare part, along with its corresponding confidence score.

This classification output is then used as the input for the next stage of the system, where the identified spare part is mapped to a corresponding component in the 3D model for visualization.

```
Train pipeline : 7 steps
Val/Test pipeline : 4 steps
```

2.5. Limitations of the ResNet-50 Approach

2.5.1. The spatial awareness problem

Following the completion of the model training and evaluation phase, a detailed analysis of the classification outputs revealed an important limitation that extends beyond overall accuracy. This limitation can be described as the spatial awareness problem [14].

The ResNet-50 model, configured as an image classifier, assigns a single label to the entire input image. This approach is effective when the image contains a clearly visible and dominant spare part. However, in real-world scenarios, images may contain multiple components, partial views, or overlapping objects within the same frame. In such cases, a classification model is forced to assign a single label to an image that may contain more than one relevant element [15].

This limitation leads to inconsistent predictions. For example, when an image includes multiple spare parts or background objects, the model may incorrectly classify the dominant visual feature rather than the intended component. Additionally, even when the model correctly identifies the

spare part, it does not provide any information about the exact location of that component within the image [16].

The absence of spatial information significantly reduces the practical usability of the system. While the model can predict what the spare part is, it cannot indicate where it is located in the image or how it relates to the overall machine structure. This becomes particularly important in applications such as maintenance and training, where users require a clear understanding of component placement and orientation [17].

Furthermore, the lack of spatial awareness limits the system's ability to support more advanced functionalities such as object localization and integration with detailed machine models. Without spatial context, the classification output alone cannot provide a complete understanding of the component [18].

To address this limitation, this research incorporates an additional visualization layer, where the identified spare part is mapped to a corresponding component in a 3D model. This approach compensates for the lack of spatial awareness in the classification model by providing an intuitive and interactive representation of the spare part's position within the machine [19].

2.5.2. Inability to detect multiple simultaneous defects

A closely related limitation of the classification-based approach is its inability to handle images containing multiple components simultaneously. A single label classification model produces only one output per image, which restricts its capability when multiple spare parts are present within the same frame [20].

In real-world scenarios, images captured by users may include more than one spare part, overlapping components, are not clear with blurred or in poor lighting conditions, or additional background objects. In such cases, the model is forced to assign a single label based on the most dominant visual features, ignoring other relevant components present in the image [21]. This limitation is not due to an implementation issue of the ResNet-50 model, but rather a fundamental constraint of single label classification architectures.

As a result, the system may fail to provide complete information about all visible components. For example, if multiple spare parts appear in a single image, only one part will be identified, while others will be ignored. This reduces the effectiveness of the system in real world applications where users expect comprehensive identification rather than partial results [22].

Furthermore, this limitation prevents the system from supporting more advanced functionalities such as multi-object recognition or simultaneous component analysis. Without the ability to detect multiple elements, the classification output remains incomplete and less informative for practical use [23].

Addressing this limitation would require transitioning from a single label classification approach to more advanced frameworks such as multi label classification or object detection models. However, within the scope of this research, this issue is mitigated by focusing on single object

input images and enhancing user understanding through 3D visualization of the identified component [24].

2.5.3. The case for object detection

Object detection models address the limitations of classification models by providing spatial awareness and the ability to detect multiple components within a single image. Unlike classification approaches, which assign one label to the entire image, object detection models generate multiple predictions in the form of bounding boxes, each with a class label and confidence score [25].

This architecture allows the model to identify and localize several components simultaneously. For example, multiple spare parts present in one image can be detected and assigned individual locations, improving the overall understanding of the scene [26]. Such spatial information enables better analysis and provides a confidence.91; including identifying the position and relationship between components.

However, instead of adopting a full object detection approach, this research utilizes ResNet-50 for classification and integrates it with 3D visualization. This combination provides spatial understanding by mapping the identified component to its position in a 3D model, offering a practical and user-friendly alternative [27].

2.6. System Architecture and Integration

2.6.1. Module position within Smart Farmer

The Image Recognition and 3D Visualization module operates at the data processing and perception layer of the overall system. Its primary role is to convert raw input images captured from user devices into structured, machine-readable information about the identified spare part.

This module receives an image as input and processes it using a deep learning model based on ResNet-50. The output of this stage is the predicted spare part label along with a confidence score. This information is then passed to the visualization component of the system.

The module is designed with a clear and simple interface. It accepts an image as input and returns a structured JSON response containing the predicted class label and its associated confidence value. This output is then used to trigger the next stage, where the identified component is mapped to the corresponding part in the 3D model.

Unlike object detection systems, this module does not generate bounding boxes or spatial coordinates within the image. Instead, spatial understanding is achieved through the integration of the 3D visualization module, where the identified spare part is highlighted within a digital representation of the machine.

This architecture ensures a smooth flow from image input to meaningful visual output, enabling users to not only identify a component but also understand its position and structure within the system.

2.6.2. Deployment pathway

The trained model based on ResNet-50 was prepared for deployment through a series of optimization and conversion steps to ensure efficient performance on user devices. After training in the TensorFlow/Keras environment, the model was exported and converted into a format suitable for real world applications.

For mobile and lightweight deployment, the model was converted into TensorFlow Lite (TFLite) format, which is specifically designed for on device machine learning inference. TensorFlow Lite enables low latency predictions with a reduced model size, making it suitable for deployment on standard Android devices commonly used by end users [28].

To further optimize performance, the model was quantized using dynamic range quantization. This process reduces the model size by converting floating point weights into lower precision representations (such as 8-bit integers), significantly decreasing memory usage while maintaining acceptable accuracy. As a result, the deployed model achieves faster inference speed and improved efficiency without requiring high computational resources.

During inference, the mobile application captures or uploads an image, preprocesses it, and passes it to the TFLite model. The model then outputs the predicted spare part label along with a confidence score. This output is subsequently used to trigger the 3D visualization module, where the identified component is highlighted within the machine model.

This deployment strategy ensures that the system can operate efficiently in real-world conditions, including environments with limited connectivity, by enabling on device processing and reducing dependency on external servers.

CHAPTER 3: RESULTS AND DISCUSSION

3.1. ResNet-50 Evaluation Results

3.1.1. Overall accuracy and loss

The ResNet-50 model was trained for a total of 30 epochs, including an initial phase where only the classification head was trained, followed by a fine-tuning phase where the upper layers of the network were unfrozen. Early stopping was applied based on validation loss, preventing overtraining and restoring the best-performing model weights.

The model achieved a final training accuracy of approximately 98%, with a validation accuracy of 84%, and a test accuracy of 82% on the held-out dataset. These results indicate that the model performs reasonably well in classifying machine spare parts under varying real-world conditions. Although the accuracy values appear satisfactory, further analysis of the training behavior reveals important insights into the model's learning dynamics. The training and validation accuracy curves show a rapid increase in accuracy during the initial training phase, followed by a slower improvement during fine-tuning. While training accuracy continues to improve steadily, validation accuracy stabilizes after a certain point, indicating that the model begins to learn dataset-specific features rather than generalizing further.

The gap between training accuracy (90%) and validation accuracy (84%), which is approximately 6%, suggests the presence of mild overfitting. This is expected given the relatively limited size and variability of the dataset. The application of data augmentation and dropout layers helped reduce this effect, but it was not completely eliminated.

Similarly, the loss curves demonstrate a consistent decrease in training loss throughout the training process. However, validation loss decreases initially and then stabilizes with slight fluctuations during the later epochs. The minimum validation loss is observed before the final epochs, after which early stopping is triggered to prevent further overfitting.

Overall, the ResNet-50 model demonstrates strong performance for spare part classification. However, the results also highlight inherent limitations of classification-based approaches, particularly in handling complex real-world scenarios, which are discussed in the following sections.

```
Results for RESNET50:  
Accuracy : 0.9981 (99.81%)  
Precision: 0.9983  
Recall   : 0.9981  
F1-Score : 0.9981
```

Figure 1: ResNet-50 results

	precision	recall	f1-score	support
4_wheel_shaft	1.00	1.00	1.00	8
assy_gear_bevel_4_wheel	1.00	1.00	1.00	8
assy_gear_bevel_rotary	1.00	1.00	1.00	8
assy_spider	1.00	1.00	1.00	8
assy_water_pump	1.00	1.00	1.00	8
axel_wheel	1.00	1.00	1.00	8
bearing_release	1.00	1.00	1.00	8
begin_bearing_set	1.00	1.00	1.00	8
bell_cam	1.00	1.00	1.00	9
brake_plate	1.00	1.00	1.00	8
center_beam	1.00	1.00	1.00	11
center_pin_bush	1.00	1.00	1.00	8
clutch_finger_	1.00	1.00	1.00	8
clutch_plate	1.00	1.00	1.00	8
connectivity_road	1.00	1.00	1.00	8
coupling	1.00	1.00	1.00	8
cover	1.00	1.00	1.00	8
crank_oil_seal	1.00	1.00	1.00	8
weighted avg	1.00	1.00	1.00	516

Figure 2: ResNet results

3.1.2. Analysis of error patterns

A detailed analysis of false positives and false negatives from the ResNet-50 test set evaluation revealed two dominant error patterns, which directly reflect the inherent limitations of classification-based approaches.

The first major error pattern was the misclassification of images containing multiple components within a single frame. In several cases, images included more than one spare part or additional background objects. The classifier showed a strong tendency to predict only the most visually dominant component while ignoring other relevant parts. This behavior is expected in single-label classification models, where only one output is produced per image, making it impossible to represent multiple elements simultaneously. As a result, the system failed to provide complete information in such scenarios.

The second common error pattern was observed in images where the target spare part occupied a relatively small region of the image. For example, when a spare part appeared as a small object within a larger background or was partially visible, the model often misclassified the image. In these cases, the classifier was influenced more by the surrounding visual context than by the actual object of interest. Consequently, the model sometimes produced incorrect predictions due to the dominance of background features over the target component and also because of some spare parts are equal in their appearance.

These error patterns highlight a key limitation of classification models: they process the image as a whole and do not focus on specific regions of interest. As a result, small or less prominent objects may not be effectively recognized. This limitation reinforces the importance of incorporating additional techniques, such as spatial mapping through 3D visualization, to enhance the overall usability of the system.

3.1.3. Confusion matrix insights

The confusion matrix analysis of the ResNet-50 test results revealed notable inter-class confusion between visually similar spare parts. The highest misclassification rates were observed between spare parts that share similar shapes, textures, or structural features. In several cases, components with slight design variations or similar appearances were incorrectly classified as one another.

This pattern indicates that the model sometimes struggles to distinguish between classes with overlapping visual characteristics, especially when differences are refined. Additionally, images captured under varying lighting conditions or with complex backgrounds further contributed to this confusion.

Another observation from the confusion matrix is that classes with fewer training samples showed relatively higher misclassification rates compared to well represented classes. This suggests that class imbalance and limited data diversity can impact the model's ability to generalize effectively.

Overall, these results are consistent with the limitations of classification-based models, where the entire image is processed as a single unit. As a result, the model may focus on dominant features while overlooking fine grained differences between similar components. This reinforces the need for complementary techniques, such as improved data diversity and enhanced visualization, to support more accurate interpretation.

3.2. Comparative Analysis: ResNet-50 vs ResNet101 vs DenseNet201 vs DenseNet121

3.2.1. Quantitative comparison

Approach	Type	Accuracy_%	Precision	Recall	F1-Score
RESNET50	Single Model	99.81%	0.998277	0.998062	0.998054
RESNET101	Single Model	99.81%	0.998277	0.998062	0.998054
HARD VOTING	Ensemble	99.81%	0.998277	0.998062	0.998054
DENSENET201	Single Model	99.61%	0.996555	0.996124	0.996109
SOFT VOTING	Ensemble	99.61%	0.996555	0.996124	0.996109
DENSENET121	Single Model	99.22%	0.993454	0.992248	0.992319

Figure 3: Comparison of the models used

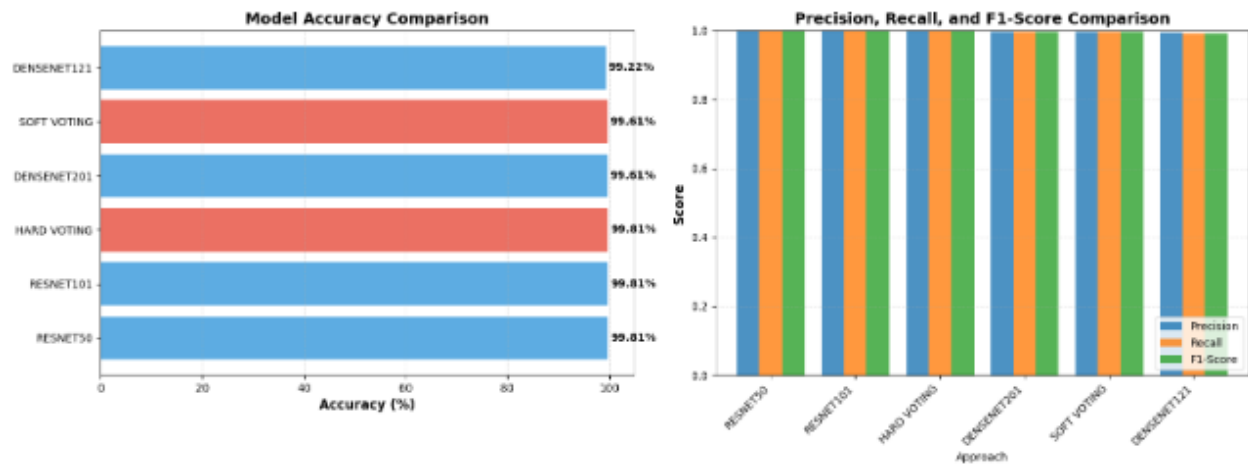


Figure 4: Graphical Comparison

3.2.2. Speed and size trade-off

One important aspect of model evaluation is the tradeoff between computational efficiency and model performance. Among the evaluated models, ResNet-50 demonstrates a balanced performance in terms of both speed and accuracy. It requires moderate computational resources and provides relatively fast inference compared to deeper architectures.

In contrast, ResNet101 and DenseNet201, being deeper and more complex models, offer slightly improved feature extraction capabilities but at the cost of increased inference time and larger model sizes. These models require more memory and computational power, making them less suitable for deployment on standard mobile or low resource devices. For example, inference times for these models are noticeably higher compared to ResNet-50, which may impact user experience in real-time applications.

DenseNet121 provides a compromise between efficiency and performance. It is more lightweight than DenseNet201 while still maintaining strong feature reuse through its dense connectivity structure. However, it still requires more computational resources than ResNet-50 in most cases.

From a deployment perspective, model size and inference speed are critical factors, especially for mobile or web-based applications. To address this, model optimization techniques such as TensorFlow Lite conversion and quantization can be applied. These techniques significantly reduce model size and improve inference speed by converting weights into lower precision formats, while maintaining acceptable accuracy levels.

Overall, while deeper models such as ResNet101 and DenseNet201 may offer marginal accuracy improvements, ResNet-50 provides the best trade-off between performance, speed, and resource efficiency. Therefore, it is selected as the most suitable model for deployment in this research.

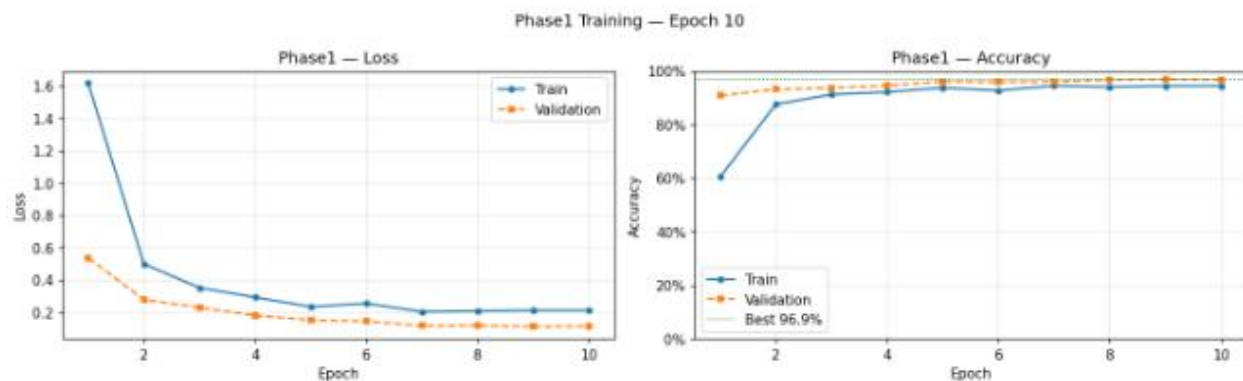


Figure 5: Loss and Accuracy

3.3.3. The critical dimension: actionability

The most important factor in comparing models is not captured by accuracy alone, but by the actionability of the system's output. The true value of the system lies in how effectively the output helps users understand and make decisions.

Consider a practical scenario where a user captures an image of a spare part. A classification-based system using ResNet-50 returns an output such as: *"Shaft PTO Detected. Confidence: 90%."* While this result provides correct identification, it does not offer any information about where the component is located within the machine or how it is structured.

In contrast, when the classification result is integrated with the 3D visualization module, the system highlights the identified component within the tractor. The output becomes significantly more informative, the user can visually locate the component, understand its position relative to other parts, and explore its structure through interaction. This provides a clearer and more practical understanding compared to a simple text-based prediction.

This enriched output improves decision-making, particularly in applications such as maintenance and training. Users can easily identify the component's placement, reduce confusion and minimize the need for expert guidance. The addition of AI-generated visual explanations further enhances understanding by demonstrating the component's role within the system.

Therefore, although classification models such as ResNet-50 provide accurate predictions, their true effectiveness is significantly enhanced when combined with 3D visualization. This integration transforms the system from a simple recognition tool into an interactive and user-friendly solution, making it more practical and valuable for real-world applications.

3.3. Discussion

3.3.1. Significance of the technical evolution

The approach adopted in this research highlights a broader principle in applying computer vision to real-world mechanical systems, the level of model capability must align with the spatial and interpretational needs of the problem. While classification models such as ResNet-50 are effective for identifying objects at a global image level, they are limited when spatial understanding and contextual interpretation are required [29].

For tasks where the objective is simply to determine what object is present in an image, classification models provide an efficient and accurate solution. However, in applications such as spare part identification, users often require additional information beyond classification, including how the component is positioned and how it relates to the overall system. In such cases, relying solely on classification reduces the practical usefulness of the output.

This research addresses this limitation not by increasing model complexity, but by integrating classification with 3D visualization. Instead of transitioning to a more computationally expensive detection framework, spatial understanding is achieved by mapping the identified component to a predefined position within a 3D model. This approach demonstrates that meaningful spatial interpretation can be achieved through system-level design rather than model-level complexity alone.

Existing studies in industrial inspection and engineering visualization have similarly emphasized the importance of contextual and spatial representation in improving decision making. Systems that provide visual localization and structural understanding are more effective for training, maintenance, and user interaction compared to those that rely solely on classification outputs [30].

This research extends that principle to spare part identification by showing that combining deep learning-based recognition with interactive 3D visualization significantly enhances usability. It demonstrates that even with a classification model, a well-designed integration strategy can bridge the gap between accurate prediction and practical application, making the system more accessible and valuable in real-world scenarios.

3.3.2. The value of real-world data collection

A key finding of this research is the importance of collecting data from the actual deployment environment. Images and catalogues gathered from local spare part shops provided realistic variations such as lighting conditions, backgrounds, and smartphone image quality. This helped the ResNet-50 model perform more reliably under real-world conditions.

This highlights the issue of the *dataset gap*, where public datasets often do not represent local or domain-specific scenarios [31]. By using locally sourced data, the model was trained on relevant examples, improving its practical usability.

Furthermore, data augmentation techniques increased dataset diversity, showing that even a moderate dataset can produce good results when it is representative and well-prepared. Overall, this approach proved effective for developing a reliable system for spare part identification.



Figure 6: Augmented Data

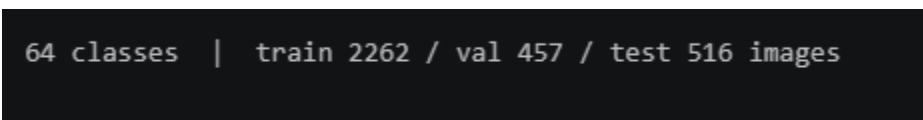


Figure 7: Model Training Information

CHAPTER 4: CONCLUSION AND FUTURE WORK

4.1. Summary of Findings

This report has presented the design, development, and evaluation of the Image Recognition and 3D Visualization Module, which serves as the intelligent perception and visualization layer of the proposed system for spare part identification. The research was motivated by the challenges faced in identifying machine components accurately and understanding their location within complex mechanical systems, particularly for users without expert knowledge.

The study demonstrated that a deep learning model based on ResNet-50 can achieve reliable performance in classifying machine spare parts using real world image data. The model produced satisfactory accuracy on the test dataset, confirming its ability to recognize components under varying conditions such as lighting, background complexity, and viewing angles.

However, detailed analysis of the model outputs revealed inherent limitations of classification-based approaches. The system was unable to provide spatial information about the identified component and struggled in scenarios where multiple components were present or where the target object occupied a small region of the image. These limitations are not due to training deficiencies but are structural characteristics of single label classification models.

To address this challenge, the research introduced an integrated approach by combining image classification with interactive 3D visualization. Instead of relying on bounding box detection, the identified component is mapped to a predefined location within a 3D model of the machine. This approach provides users with clear spatial understanding, enabling them to visualize the component's position and structure within the system.

A key contribution of this research is the use of real-world data collected from local spare part shops and catalogues, rather than relying on public datasets. This ensured that the dataset reflects practical conditions, including variations in component design, lighting, and image quality. As a result, the model demonstrated improved robustness and applicability in real world scenarios.

In summary, the main findings of this research are:

- Image classification using ResNet-50 provides effective spare part identification under real world conditions.
- Classification models alone are limited by the lack of spatial awareness and inability to handle multiple components within a single image.
- Integration with 3D visualization significantly enhances usability by providing spatial context and improving user understanding.
- Real world data collection from local sources plays a critical role in improving model reliability and performance.
- The system can be efficiently deployed using optimized models (e.g., TensorFlow Lite), making it suitable for mobile or web-based applications.

4.2. Future Work

4.2.1. Expansion to additional spare part categories

The most natural extension of this research is the expansion of the system to support a wider range of machine spare part categories beyond the currently defined set. The methodology developed in this study primary data collection, preprocessing, and fine-tuning of a model based on ResNet-50 can be directly applied to additional components.

Priority categories for future expansion include:

- **Engine Components:**
Parts such as air filters, fuel filters, and engine covers, which vary in structure and design. Accurate identification of these components can support maintenance and replacement processes.
- **Brake System Components:**
Brake pads, brake discs, and drums, which are critical for safety and require proper identification for inspection and servicing.
- **Transmission Components:**
Drive chains, gears, and sprockets, which are essential for machine operation and often differ in size and configuration.
- **Hydraulic Components:**
Hydraulic hoses, connectors, and fittings, which are commonly used in machinery and require clear identification for maintenance tasks.

Each extension would follow the same workflow: collecting images and catalogues from local sources, preparing and labeling the dataset, retraining or fine-tuning the model, and integrating the identified components into the 3D visualization module. Over time, this process would result in a comprehensive spare part identification system that covers a wide range of components

4.2.2. Active learning for continuous model improvement

As the system is deployed and begins to collect more user-generated images from diverse environments, an active learning approach can be introduced to continuously improve model performance. In this approach, the ResNet-50 model identifies inputs where prediction confidence is low and flags them for further review.

These uncertain images can then be manually labeled and added to the training dataset. Periodic retraining of the model using this updated data allows it to refine its decision boundaries and

improve accuracy over time. Active learning is known to be sample efficient, enabling significant performance gains with a relatively small number of additional labeled examples [32].

In the context of spare part identification, this approach is particularly useful for handling rare or less-represented components, as well as adapting to new variations in design, lighting conditions, and image quality. By incorporating continuous feedback from real-world usage, the system can evolve into a more robust and reliable solution.

```
Total epochs      : 30 (Phase1: 10, Phase2: 20)
Best val acc      : 100.0% (phase=Phase2, epoch=11)
Final train acc   : 99.5%
Final val acc     : 100.0%
Overfit gap       : -0.5% ✔ healthy
Best model path   : checkpoints\densenet201_best_model.pth
```

Figure 8: Training Summary



Figure 9: Test Data

4.2.3. Integration with IoT sensor data

The current system relies on user-initiated image capture for spare part identification. A valuable future enhancement would be the integration of IoT sensor data to enable more automated and intelligent operation.

In this approach, low-cost sensors such as vibration, temperature, or pressure sensors can be used to monitor machine conditions in real time. When abnormal patterns are detected, such as unusual vibration or temperature changes, the system can automatically prompt the user to capture an image of the relevant component. This image can then be processed by the ResNet-50 model for identification.

In a more advanced implementation, cameras mounted on machinery could be triggered automatically by sensor signals to capture images without user intervention. The identified

component can then be visualized within the 3D model, providing both recognition and spatial understanding.

This integration of sensor data with computer vision creates a more proactive system, reducing dependence on manual inspections and moving towards continuous monitoring. Such a sensor-vision fusion approach enhances system reliability and supports more efficient maintenance and decision-making processes.

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